

Devesh Maheshwari

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EDUCATION

Purdue University – West Lafayette
Bachelor of Science – Computer Science & Electrical Engineering

December 2026
GPA: 3.98 / 4.00

RELEVANT COURSEWORK

Data Structures and Algorithms, Analysis of Algorithms, Data Mining & ML, Artificial Intelligence, C Programming, Python for Data Science, Computer Architecture, Systems Programming, Competitive Programming, Robotics

TECHNICAL SKILLS

Programming Languages: Python, Java, C, C++, JavaScript/TypeScript, SQL, Assembly x86-64, Lex/Yacc, YAML
Frameworks & AI/ML: FastAPI, React, Node.js, HTML/CSS, REST APIs, LLM APIs, PyTorch, scikit-learn, RAG (FAISS)
Tools & Infra: Git, GitHub, GitLab, Docker, AWS, MySQL, PostgreSQL, MongoDB, Linux

EXPERIENCE

AI / ML Intern – StudioX-AI Jun 2025 – Aug 2025

- Automated DevOps and customer support with AI workflows, boosting internal efficiency and saving 5-8 hrs weekly.
- Integrated GitLab, Docker APIs with Claude via YAML for CI/CD and user management, sanity checks, and image builds.
- Packed integrations into DevOps chatbot enabling 10+ operations through conversational interface, reducing manual effort.
- Streamlined customer email intake with Claude, semantically analyzing ~50 emails/month to generate tickets.

Applied AI Intern – Sopra Steria May 2024 – Jul 2024

- Delivered RAG search over employee PDFs, reducing lookup time and generating grounded answers in under 5s.
- Synthesized ingestion pipeline by parsing PDFs, chunking ~800 tokens, embedding with OpenAI, indexing in FAISS db.
- Connected FAISS retrieval with GPT-4 to synthesize answers from top passages, improving response accuracy to 90%.

AI Developer – BytEdge Inc. May 2023 – Jul 2023

- Enabled natural language search on company platform, cutting lookup from minutes to under 10s for 200+ daily queries.
- Fine-tuned GPT-3 with ~300 prompts to generate SQL tailored to company schema, retrieving records from complex menus.
- Implemented GPT-3 outputs with summarization pipeline, presenting results in natural language within search bar interface.

App Developer – Equinox Solutions LLP May 2022 – Aug 2022

- Created SeedTrace Android app for supply chain transparency in organic food, giving 700+ consumers verifiable details.
- Programmed QR code scanning with ZXing in Java/JS to show farm origin, harvest date, warehouse location, certifications.
- Linked React Native frontend with REST APIs to deliver product data for 100+ items directly within mobile interface.

PROJECTS

Lunar Lander Game – FPGA and Verilog Project Jul 2025 – Aug 2025

- Engineered modular Verilog components (ALU, memory, control, display units) for FPGA lunar lander simulation at 100 Hz.
- Devised 16-bit BCD arithmetic with adders/subtractors, enabling single-cycle updates for altitude, velocity, fuel, thrust.
- Verified implementation on hardware with 100+ test cases, driving 7-segment displays and LEDs for real-time user feedback.

Shell Project – Advanced Custom Bash Shell Jan 2024 – Mar 2024

- Constructed custom Unix-like shell in C/C++ with Lex/Yacc, supporting subshells, loops, wildcards, pipes, I/O redirection.
- Configured modular process, environment manager with 10+ built-in commands & extensible scripting language features.
- Achieved broad feature coverage across 50+ detailed test cases, ensuring correctness of parsing, execution, and redirection.

Simple C Compiler – Compiler Design Project Oct 2023 – Dec 2023

- Developed compiler from scratch for a subset of C using Lex and Yacc, translating source code into x86-64 assembly.
- Programmed stack-based expression evaluation, control-flow constructs, and function call handling with register optimization.
- Formulated symbol tables and scope management supporting global, local, and nested identifiers across 40+ test programs.

Autonomous Motorsports Purdue – Computer Vision Project Sep 2022 – Mar 2023

- Used MobileSAM CNN model to detect track boundaries and segment images on-device, improving navigation speed by 70%.
- Incorporated ZED camera for accurate position tracking & racing line computation, enabling 30 FPS real-time analysis.
- Calibrated sensor inputs, synchronized vision pipeline with control loop, ensuring stable autonomous steering at high speeds up to 40 mph.